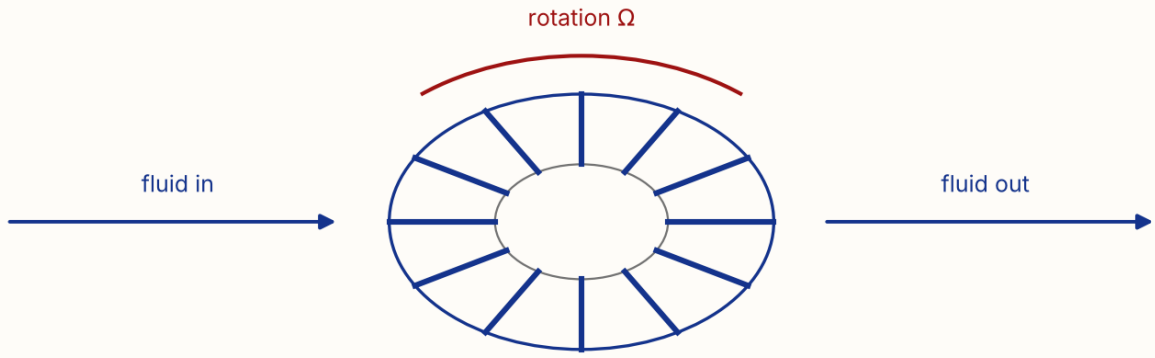
UNIT 1: Introduction and Classification

Turbo Machinery | Handwritten-style exam notes | 27 Oct 2026

Syllabus covered: Axial, radial and mixed flow machines; equations of motion in rotating frame; Coriolis and centrifugal effects; momentum and energy equation; Euler work; examples.

1. What is a turbomachine?

Rotor and stator: fluid in, fluid out, shaft work



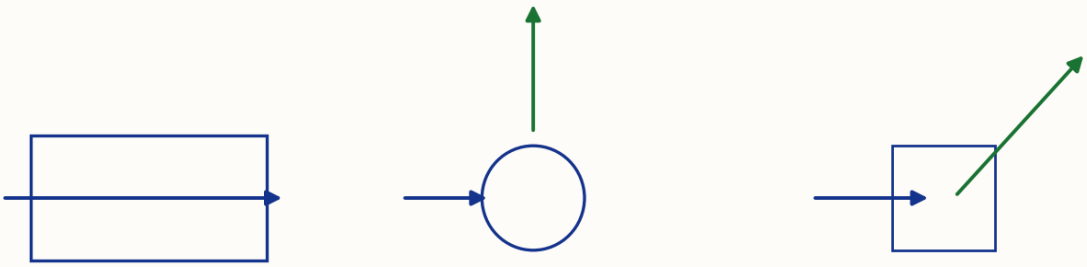
Turbine: fluid gives work to shaft. Compressor/pump: shaft gives work to fluid.

A turbomachine is a device in which **energy is transferred continuously between a rotating shaft and a flowing fluid** through a row of blades. The rotor blades change the fluid's angular momentum, and this changes its stagnation enthalpy.

- **Power producing** (fluid does work on the shaft): steam, gas and hydro turbines, wind turbines.
- **Power absorbing** (shaft does work on the fluid): pumps, fans, blowers, compressors.
- Turbomachines are preferred when large flow rates, high speed and compact size are needed.

2. Classification by flow direction

Axial, radial and mixed flow paths



The main classification is the direction of flow through the rotor:

- **Axial flow:** flow stays parallel to the shaft axis. Used in gas turbines and axial compressors. Large mass flow, moderate pressure rise per stage.

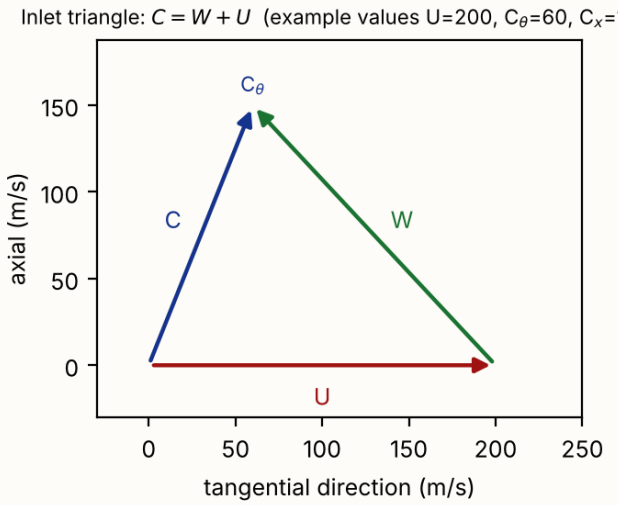
- **Radial (centrifugal) flow:** flow enters axially and leaves radially. High pressure rise per stage, wider operating range, but larger frontal area.
- **Mixed flow:** entry is axial, exit has both radial and axial components. Used in mixed-flow pumps and some turbochargers.

Feature	Axial flow	Radial flow	Mixed flow
Flow path	Parallel to axis	Radial outward/inward	Between axial and radial
Pressure rise / stage	Low to moderate	High	Moderate
Mass flow	High	Low to moderate	Moderate
Frontal area	Small	Large	Medium
Typical use	Jet engines, big fans	Small gas turbines, pumps	Mixed pumps, turbochargers

Other classifications: by energy direction (turbine vs pump/compressor), and by fluid type. Incompressible machines (pumps, hydro turbines) and compressible machines (compressors, gas and steam turbines) are analysed differently because density changes in the second group.

3. Velocity triangles (basic)

Velocity triangle: C = W + U



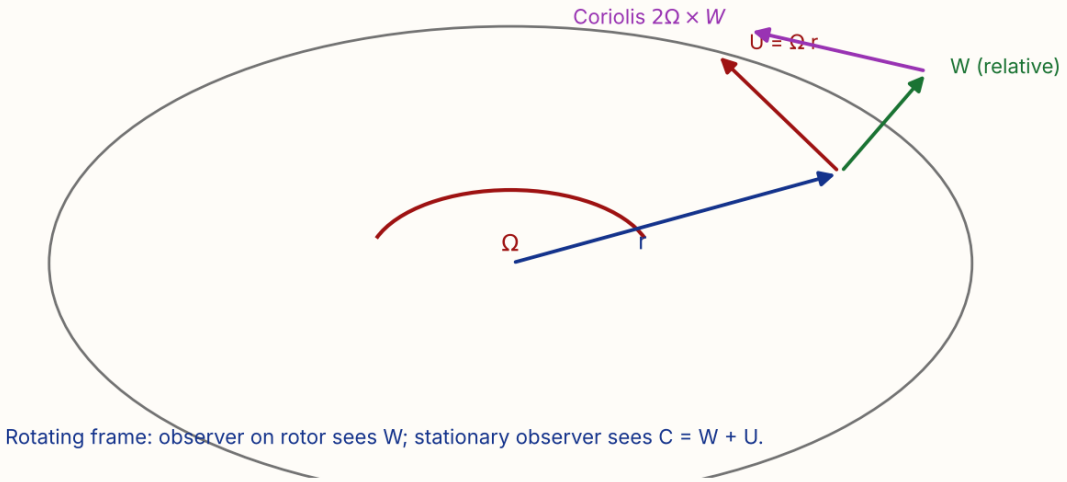
At any radius, the fluid velocity is shown as a vector triangle. Let U be the blade speed, C the absolute velocity, W the relative velocity (seen by an observer on the rotor), and alpha and beta the absolute and relative flow angles.

C = W + U where U = Ω r (C_x = axial, C_θ = tangential components)

Velocity components are used most often: C_x (axial, carries the mass flow) and C_θ (tangential, carries the angular momentum that gives work).

4. Equations of motion in a rotating frame

Rotating frame: W, U and the Coriolis direction



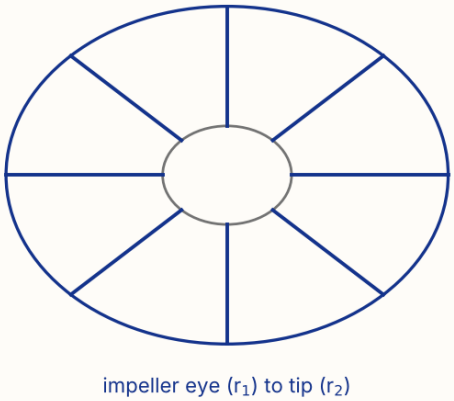
Take a rotor turning at angular velocity Ω. An observer on the rotor measures relative velocity W and relative acceleration. The absolute acceleration is related to the relative one by the extra terms of the rotating frame:

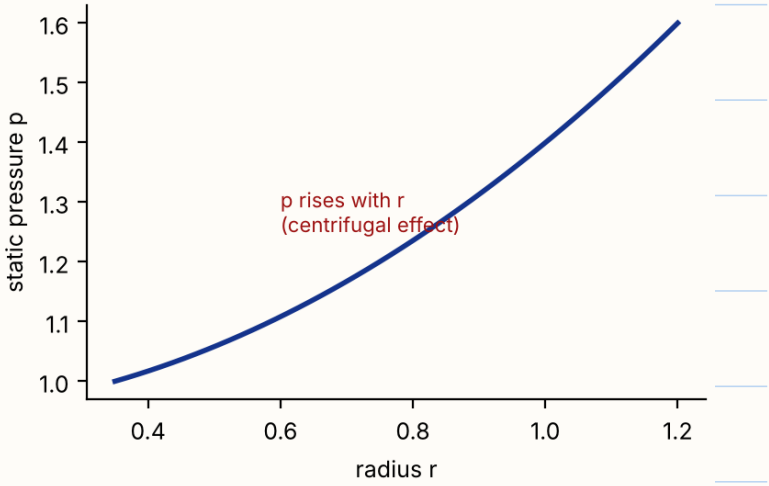
$$\mathbf{a_{abs} = a_{rel} + 2\,(\Omega \times W) + \Omega \times (\Omega \times r)}$$

- **Coriolis acceleration** $2(\Omega \times W)$: acts perpendicular to W, magnitude $2\Omega W$. It deflects the relative flow inside the rotor passage.
- **Centrifugal acceleration** $\Omega \times (\Omega \times r)$: acts radially outward, magnitude $\Omega^2 r$. It pumps fluid outward in radial machines.
- The rotating-frame momentum equation therefore has two extra source terms: $-2\rho(\Omega \times W)$ and $-\rho\Omega \times (\Omega \times r)$.

5. Effects of Coriolis and centrifugal forces

Centrifugal pressure rise in a radial impeller





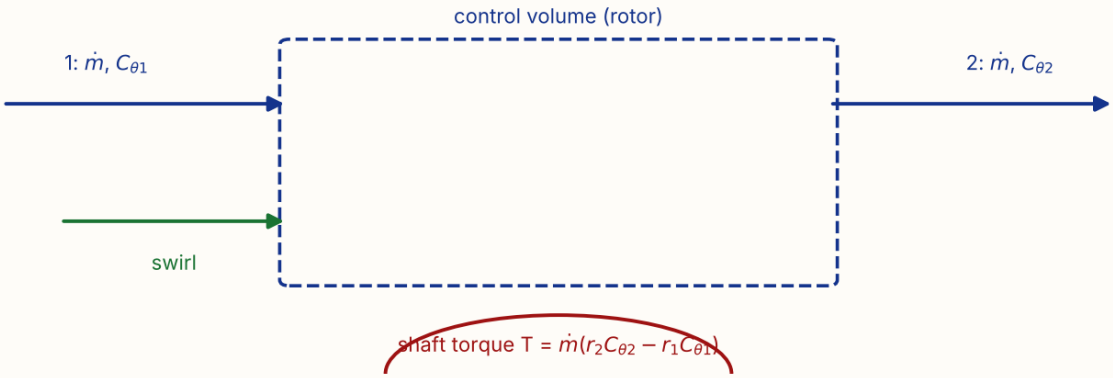
Centrifugal effect: in a radial machine the fluid is thrown outward. Even without diffusion the static pressure rises from inlet to outlet. Part of the rise comes from the rotor (centrifugal head) and part from relative velocity change:

$$p_2 - p_1 = \rho(U_2^2 - U_1^2)/2 + \rho(W_1^2 - W_2^2)/2$$

Coriolis effect: the Coriolis force acts across the relative flow, so the velocity is not uniform across the passage. Low-momentum fluid collects on one side. This non-uniform velocity causes slip (exit flow angle differs from blade angle) and can cause reversed eddies at off-design flow rates. Both are studied in Unit 4.

6. Momentum and moment of momentum equation

Control volume: torque from the change in swirl



For steady flow through a control volume, mass is conserved and the moment of momentum changes only because of torque on the fluid.

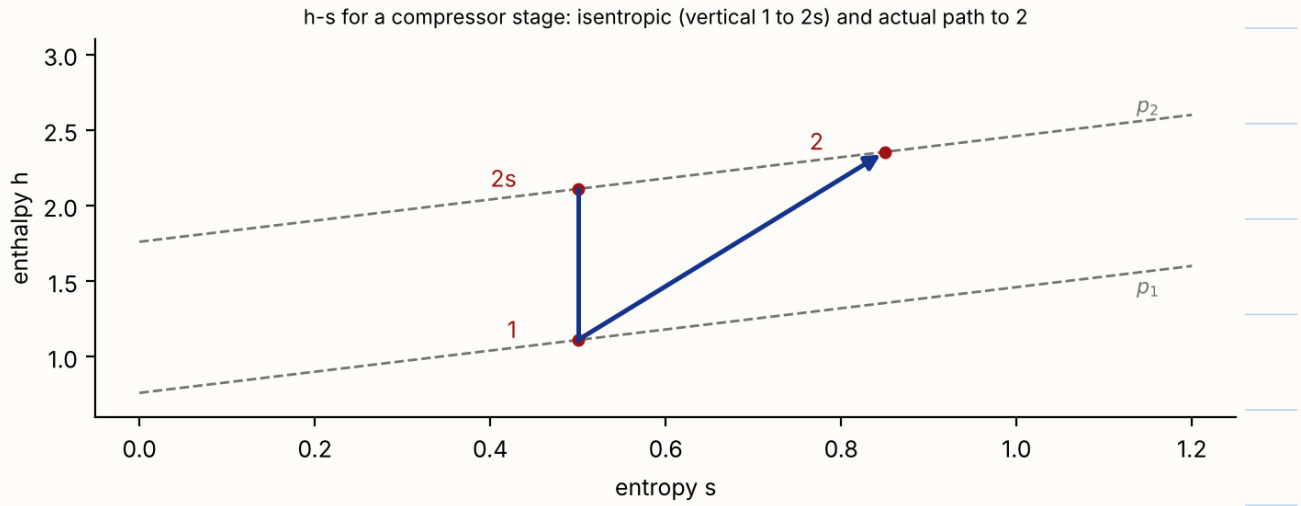
The moment-of-momentum (Euler turbomachine) equation is:

$$\mathbf{T = \dot{m}\,(r_2\,C_{\theta 2} - r_1\,C_{\theta 1}) \quad \Rightarrow \quad Power\ P = T\,\Omega = \dot{m}\,(U_2\,C_{\theta 2} - U_1\,C_{\theta 1})}$$

Here $\dot{m}$ is mass flow rate. Sign convention: C_θ is positive in the direction of blade rotation. For a compressor the power is input; for a turbine the sign of the result reverses (the fluid gives up energy).

7. Energy equation and Euler work

h-s diagram for a compressor stage



For steady, adiabatic flow with shaft work w, the energy equation is $h_{01} + w = h_{02}$. Dividing the power by mass flow gives the specific work:

$w = h_{02} - h_{01} = U_2 C_{\theta 2} - U_1 C_{\theta 1}$ (Euler work equation)

The Euler head is $H_E = w / g$ in metres of fluid. This is the ideal (no loss) head. Work done by the rotor depends only on the change in angular momentum, not on the detailed blade shape. This is why velocity triangles are the key tool.

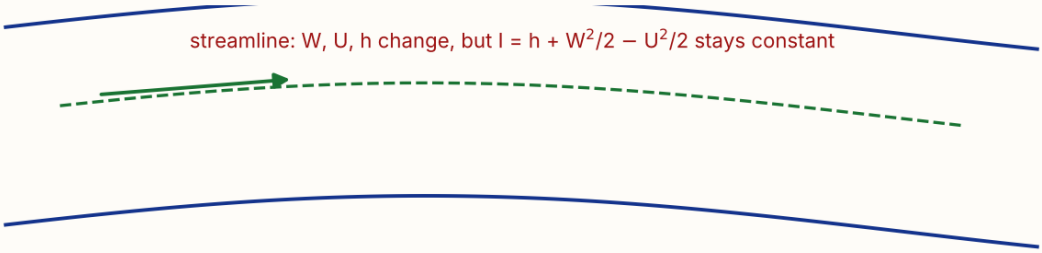
Thermodynamic link: for any element of fluid, $T\,ds = dh - dp/\rho$. In a compressor the actual work is larger than the isentropic work because of losses:

$$\eta_c = (T_{02s} - T_{01}) / (T_{02} - T_{01}) \qquad \eta_t = (T_{01} - T_{02}) / (T_{01} - T_{02s})$$

For a turbine, the actual drop is smaller than the isentropic drop, so efficiency is written with the actual drop in the numerator.

8. Rothalpy (useful in rotor analysis)

Rotor passage along a streamline



adiabatic rotor passage (schematic)

In adiabatic flow through a rotor, the rothalpy is conserved along a streamline, where it is defined as:

$$I = h + W^2/2 - U^2/2 = \text{constant along a streamline}$$

(adiabatic, no work in the relative frame)

Rothalpy shows that a change in W at constant h is linked to a change in U. It is often used to find the exit state of a rotor without solving the

whole flow field.

9. Worked examples

Working space for the examples (cover the solution and try first)



Example: *A centrifugal compressor impeller has U_2 = 350 m/s, $C_{\theta 2}$ = 250 m/s, $C_{\theta 1}$ = 0 and $\dot{m}$ = 5 kg/s. Find the specific work and the power.*

Solution: $w = U_2 C_{\theta 2} = 350 \times 250 = 87.5$ kJ/kg. Power = $\dot{m}\, w = 5 \times 87.5 = 437.5$ kW.

Example: *Air enters an axial compressor stage at T_{01} = 300 K. The stage work is 40 kJ/kg and η_c = 0.85. Find T_{02} and the isentropic work.*

Solution: c_p = 1.005 kJ/kgK, so $T_{02} - T_{01}$ = 40 / 1.005 = 39.8 K and T_{02} = 339.8 K. Isentropic work = $\eta_c \times w$ = 0.85 × 40 = 34 kJ/kg.

Example: *A pump moves 0.05 m³/s of water with a head of 30 m. Find the water power.*

Solution: P = ρ g Q H = 1000 × 9.81 × 0.05 × 30 = 14.7 kW.

10. Exam quick points

- Axial = along axis, radial = outward, mixed = both.
- Euler work: $w = U_2C_{\theta 2} - U_1C_{\theta 1}$. Remember it is the change in angular momentum.
- Coriolis causes non-uniform relative flow and slip. Centrifugal force causes static pressure rise in radial machines.
- Always state the sign convention for swirl (C_θ is positive along rotation).

11. Practice questions

Q1. Classify a jet engine compressor and a hydro-turbine by flow direction. Give one reason for each choice.

Q2. Derive the Euler work equation from the moment-of-momentum equation.

Q3. A pump has U_2 = 25 m/s and $C_{\theta 2}$ = 18 m/s with no pre-whirl. Find the Euler head (g = 9.81 m/s²).

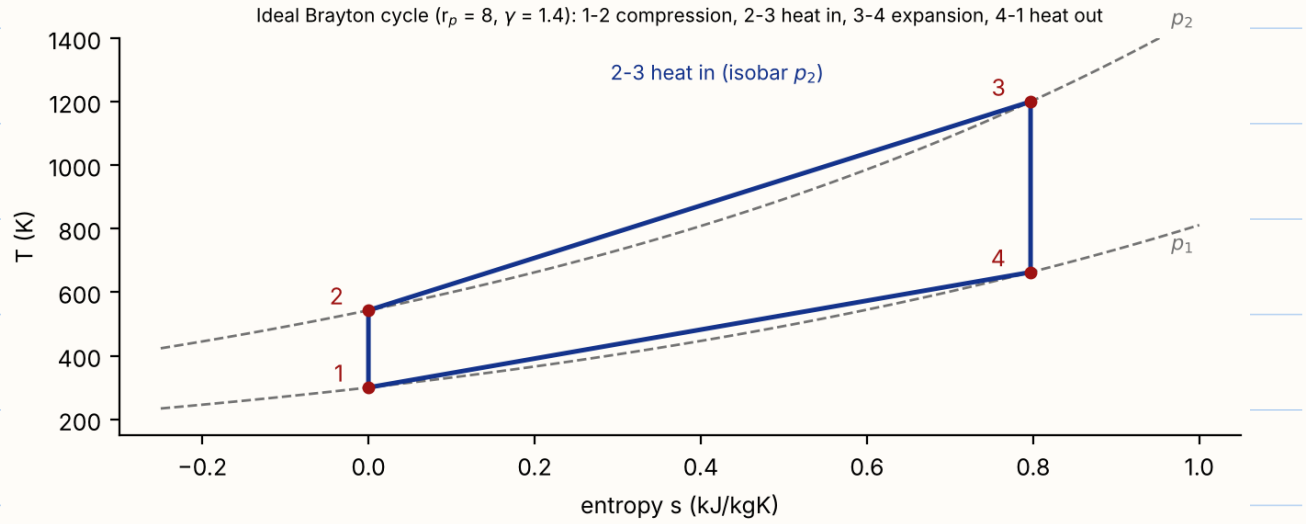
UNIT 2: Gas Turbine Cycle

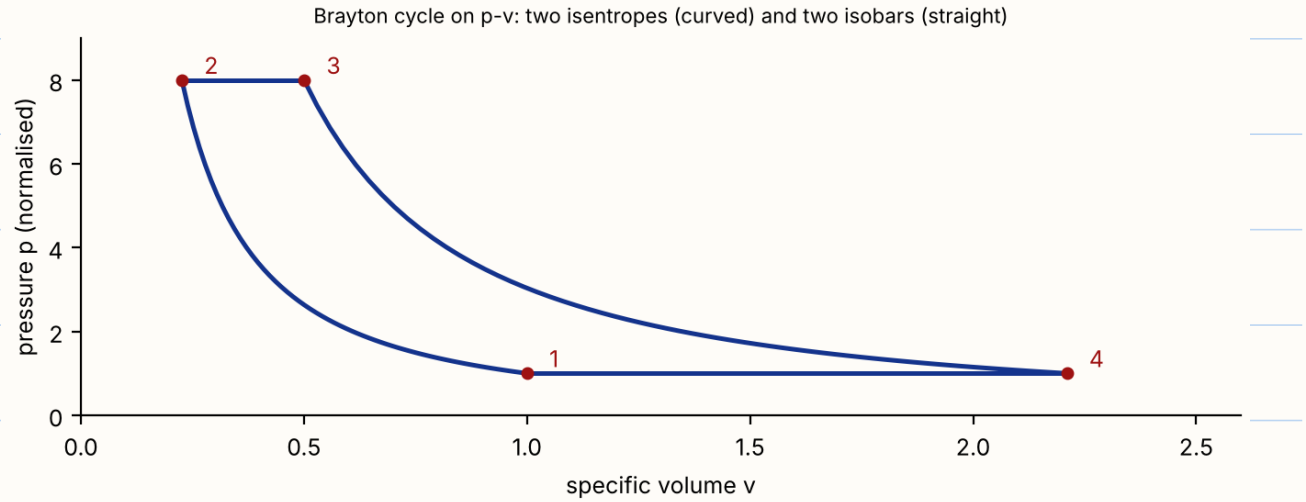
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Syllabus covered: Brayton cycle; regenerative cycle; reheat; inter-cooling; turboprop, turbojet and turbofan engines; thrust augmentation; examples.

1. The ideal Brayton cycle

Brayton cycle on T-s and p-v





The Brayton cycle is the ideal cycle for gas turbines. It has four processes:

- 1–2: isentropic compression in the compressor.
- 2–3: constant-pressure heat addition in the combustor.
- 3–4: isentropic expansion in the turbine.
- 4–1: constant-pressure heat rejection (exhaust to atmosphere).

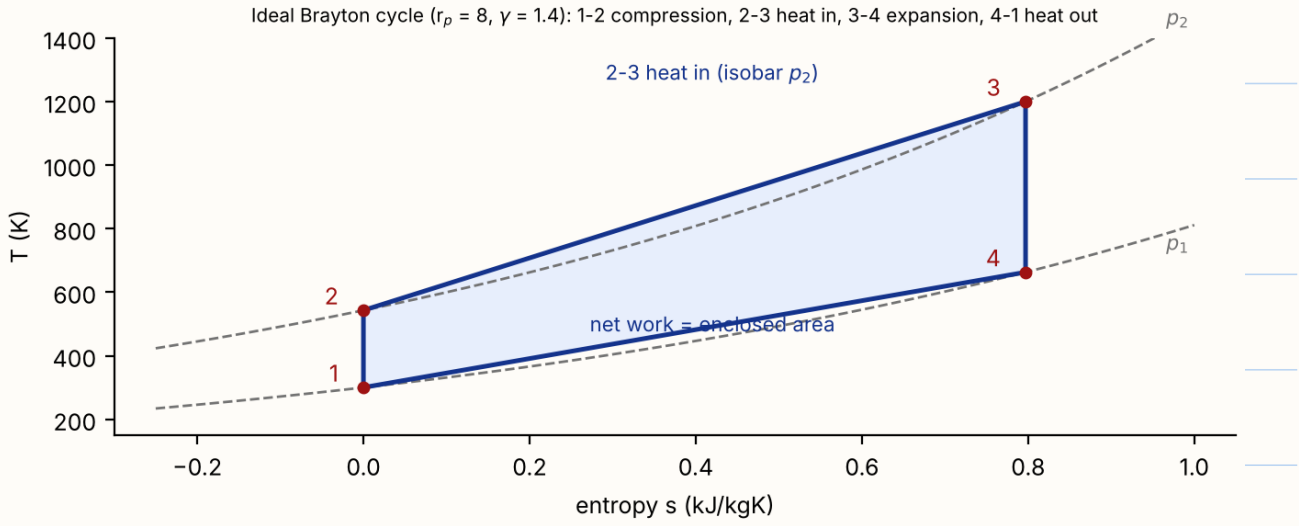
On a T-s diagram the cycle is two isobars ($p_2 = p_3$, $p_1 = p_4$) joined by two isentropes. On a p-v diagram it is a closed loop. Net work equals the area enclosed.

Pressure ratio $r_p = p_2 / p_1$ ⇒ $\eta_{th} = 1 - 1 / r_p^{(\gamma-1)/\gamma}$

Important result: for the ideal cycle, efficiency depends only on the pressure ratio. It does not depend on the turbine inlet temperature.

2. Work and back-work ratio

Work as areas on the T-s diagram



$w_c = c_p(T_2 - T_1) = c_p T_1(r_p^{(\gamma-1)/\gamma} - 1)$ $w_t = c_p(T_3 - T_4)$

$w_{net} = w_t - w_c$

The **back-work ratio** is w_c / w_t. Gas turbines have a large back-work ratio (often 40–80%) because compression consumes most of the turbine work. This is why improving compressor and turbine efficiency matters so much.

Maximum net work occurs at an optimum pressure ratio. For a given maximum and minimum temperature (T_3 and T_1):

$$r_{p,opt} = (T_3 / T_1)^{\gamma / 2(\gamma-1)}$$

Example: *For $r_p = 8$ and $\gamma = 1.4$, find the ideal Brayton efficiency.*

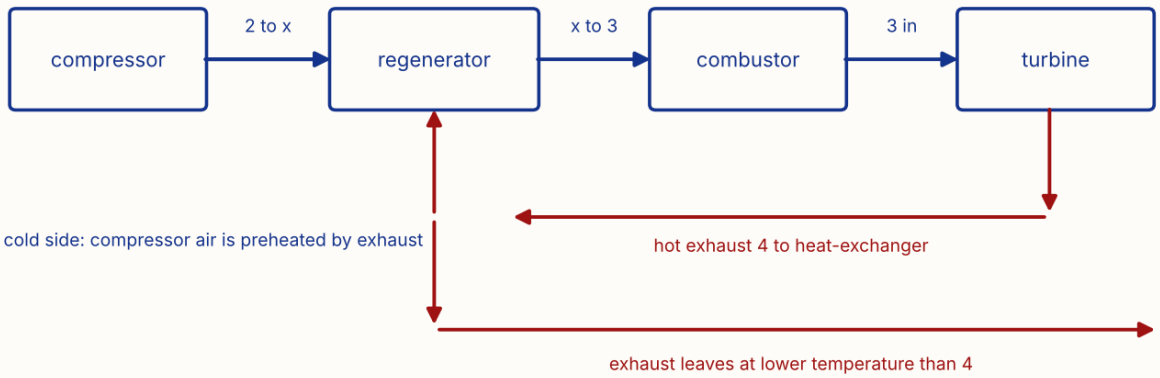
Solution: $r_p^{(\gamma-1)/\gamma} = 8^{0.2857} = 1.811$, so $\eta_{th} = 1 - 1/1.811 = 0.448$, i.e. about 44.8%.

3. Real cycle losses

Real components are not isentropic. The compressor efficiency and turbine efficiency reduce the net work. Combustor pressure loss (typically 3–6% of inlet pressure) and the mechanical losses also matter. Turbine inlet temperature (TIT) is limited by materials and cooling, so TIT is the main design variable for improving output.

4. Regenerative cycle

Regenerative cycle layout



A regenerator (heat exchanger) uses hot exhaust gas to preheat the compressor delivery air before combustion. Heat added in the combustor is reduced, so the cycle efficiency rises.

Effectiveness $\varepsilon = (T_x - T_2) / (T_4 - T_2)$ $\quad$ **$q_{in} = c_p(T_3 - T_x)$**

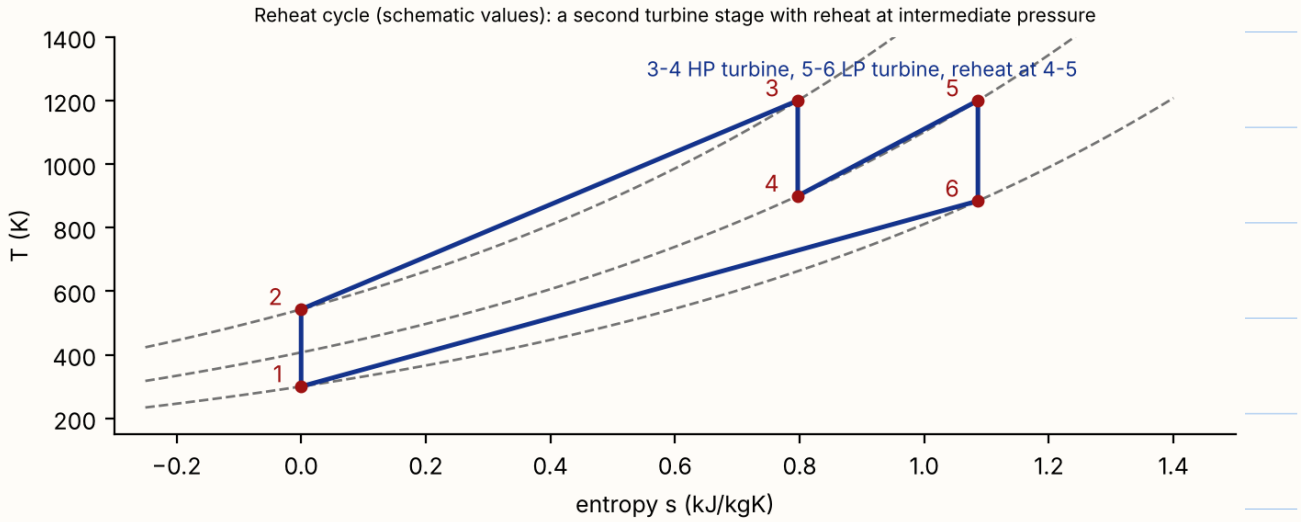
Here T_x is the air temperature leaving the regenerator on the compressor side. An ideal regenerator has $\varepsilon = 1$ and $T_x = T_4$.

- Regeneration is most useful at **low pressure ratios**, where the exhaust is still hot compared with the compressor delivery.
- At high pressure ratios the compressor outlet is already hotter than the exhaust, so the regenerator gives no benefit.

- The regenerator adds pressure loss and cost, so it is a trade-off with the efficiency gain.

5. Reheat and inter-cooling

Reheat cycle on T-s (schematic values)



Inter-cooling: compression is split into two stages with a cooler between them. Cooling reduces the specific volume of air, so less work is needed for the second stage. For two stages with the same pressure ratio per stage, the minimum compressor work occurs when:

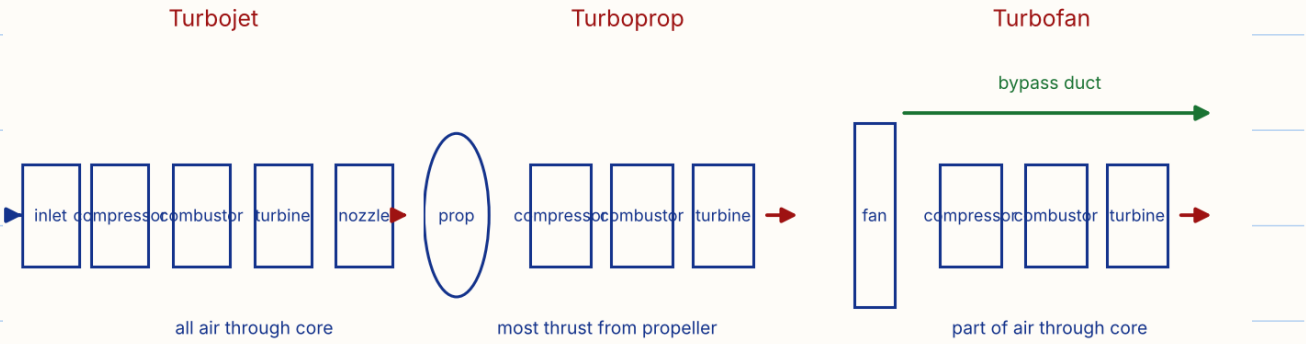
Intermediate pressure $p_i = \sqrt{(p_1\ p_2)}$ (equal pressure ratio per stage)

Reheat: expansion is split into two turbines with a second combustor between them. The gas is reheated before the second turbine. Turbine

work increases, so net work increases for the same compressor work. Intercooling and reheat together increase the overall net work but add complexity. Regeneration, intercooling and reheat are often combined to get high efficiency.

6. Aero engines: turbojet

Turbojet, turboprop and turbofan



In a turbojet, all air passes through the core. The gas leaves the nozzle at high velocity and the reaction gives thrust. The main thrust equation is:

$$F = \dot{m}\,(V_e - V_0) + (p_e - p_0)\,A_e$$

Here V_0 is flight speed, V_e is the jet exit velocity, p_e and p_0 are the exit and ambient pressures, and A_e is the nozzle exit area. If the nozzle is

fully expanded ($p_e = p_0$), only the momentum term remains.

Propulsive efficiency is the ratio of useful propulsive power to the rate of kinetic energy given to the jet:

$$\eta_p = 2\,V_0 / (V_e + V_0) \qquad \eta_0 = \eta_{th} \times \eta_p$$

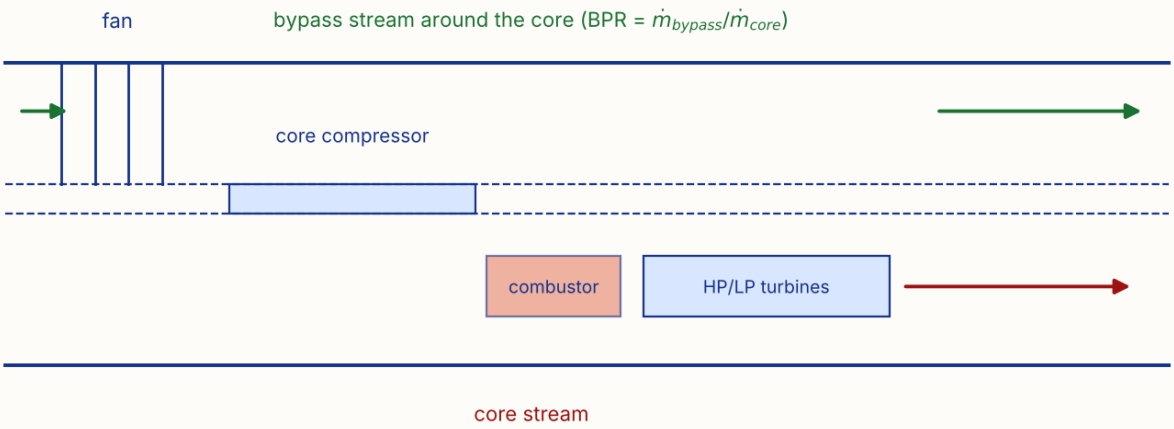
Propulsive efficiency is high when V_e is close to V_0. The turbojet has a high jet velocity, so it is efficient only at high speed.

7. Turboprop

In a turboprop, the gas generator drives a propeller through a reduction gearbox. Most of the thrust comes from the propeller, which moves a large mass of air at low velocity. This gives high propulsive efficiency at low to medium speeds (up to about 0.8 Mach), and it is ideal for regional transport.

8. Turbofan

Turbofan cross-section



A fan at the front pushes air through a bypass duct around the core.

The bypass ratio is:

$$\mathbf{BPR = \dot{m}_{bypass} / \dot{m}_{core}}$$

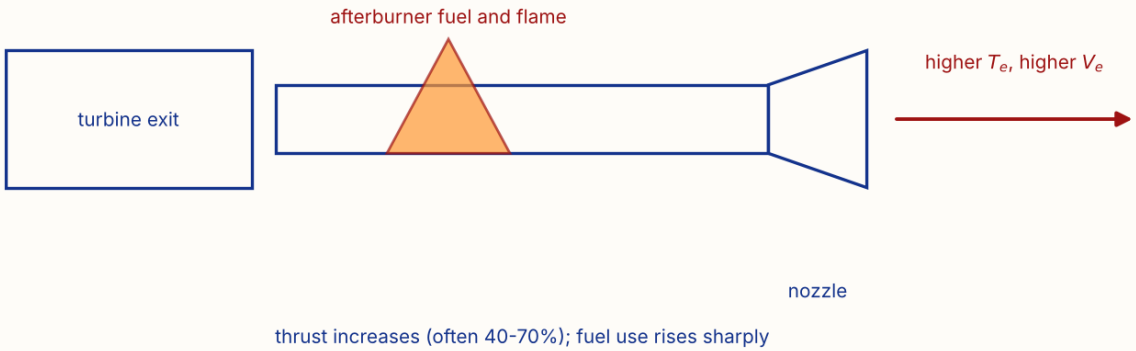
- Large BPR (high-bypass turbofan): low jet velocity, high propulsive efficiency, lower fuel consumption and lower noise.
- Low BPR (low-bypass turbofan): higher jet velocity, used for supersonic and military applications.
- Turbofans cover most civil aviation today because they combine the efficiency of the propeller with the speed of the turbojet.

Engine	Air through core?	Main thrust source	Best speed range

Turbojet	All air	High-velocity jet	High speed
Turboprop	All air	Propeller	Low to medium speed
Turbofan	Part of air	Fan + jet	Medium to high speed

9. Thrust augmentation (afterburner / reheat)

Afterburner in the jet pipe



thrust increases (often 40–70%); fuel use rises sharply

Fuel is injected into the hot jet pipe downstream of the turbine and burned. This raises the exit temperature and velocity, so the thrust increases (often by 40–70%). The cost is a large increase in specific fuel consumption, so afterburners are used mainly for take-off and combat. Water or methanol injection into the compressor is another way to boost thrust for a short time.

10. Worked examples

Example: *A turbojet has $\dot{m}$ = 50 kg/s, flight speed V_0 = 250 m/s, exit velocity V_e = 600 m/s and $p_e = p_0$. Find the thrust and propulsive efficiency.*

Solution: F = 50 × (600 − 250) = 17.5 kN. η_p = 2 × 250 / (600 + 250) = 500 / 850 = 0.588.

Example: *A regenerator has ε = 0.8, T_2 = 500 K and T_4 = 800 K. Find T_x.*

Solution: $T_x = T_2 + \varepsilon (T_4 - T_2)$ = 500 + 0.8 × 300 = 740 K.

11. Exam quick points

- Ideal Brayton efficiency depends only on pressure ratio: $\eta = 1 - 1/r_p^{(\gamma-1)/\gamma}$.
- Back-work ratio of a gas turbine is high, so compressor efficiency is critical.
- Regenerator helps at low pressure ratio. Intercooling uses equal pressure ratios for minimum work.
- Thrust = $\dot{m}(V_e - V_0) + (p_e - p_0)A_e$. High BPR turbofan gives high η_p.

12. Practice questions

Q1. Draw the T-s diagram of the ideal Brayton cycle and label all four processes.

Q2. Explain why regeneration is less useful at high pressure ratio.

Q3. A turbofan has BPR = 5, core flow 40 kg/s and V_0 = 230 m/s. Explain qualitatively how the thrust is shared between the fan and core.

UNIT 3: Similarity Analysis and Cascade Analysis

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Syllabus covered: Similarity rules, specific speed, Cordier diagram and examples; two-dimensional cascade theory, lift and drag, blade efficiency, loss estimation, compressor and turbine cascades, blade geometry, examples.

1. Why similarity?

Testing full-size machines is expensive. Similarity analysis lets us predict the performance of one machine from another that is geometrically and dynamically similar. Two machines are similar when all their dimensionless groups are the same.

- Geometric similarity: all lengths are in a fixed ratio.
- Kinematic similarity: velocity triangles are similar at corresponding points.
- Dynamic similarity: Reynolds number, Mach number and ratio of specific heats match (for compressible flow).

2. Dimensionless groups

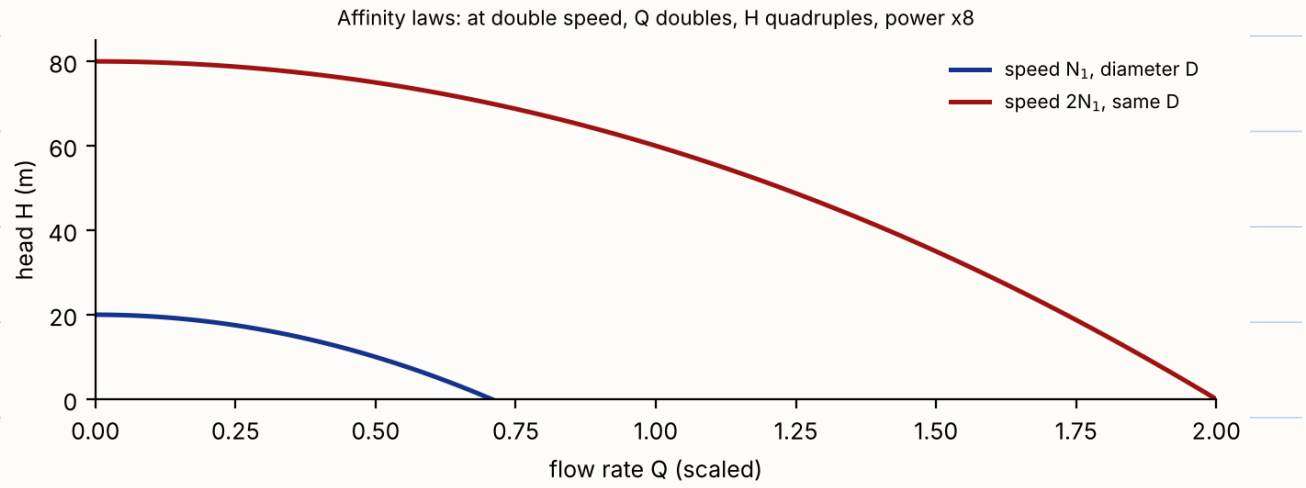
The main groups used in turbomachinery are:

Flow coefficient $\varphi = C_x / U$ Head (stage loading) coefficient $\psi = gH / U^2 = \Delta h_0 / U^2$

Power coefficient $\lambda = P / (\rho N^3 D^5)$ Reynolds number $Re = \rho U c / \mu$ Mach number $M = U/a$

3. Affinity (similarity) laws

Affinity laws: head against flow at two speeds



For the same machine, or two geometrically similar machines, with constant density, the performance changes with speed N and diameter D as:

$$Q_2/Q_1 = (N_2/N_1)(D_2/D_1)^3 \quad H_2/H_1 = (N_2/N_1)^2(D_2/D_1)^2 \quad P_2/P_1 = (N_2/N_1)^3(D_2/D_1)^5$$

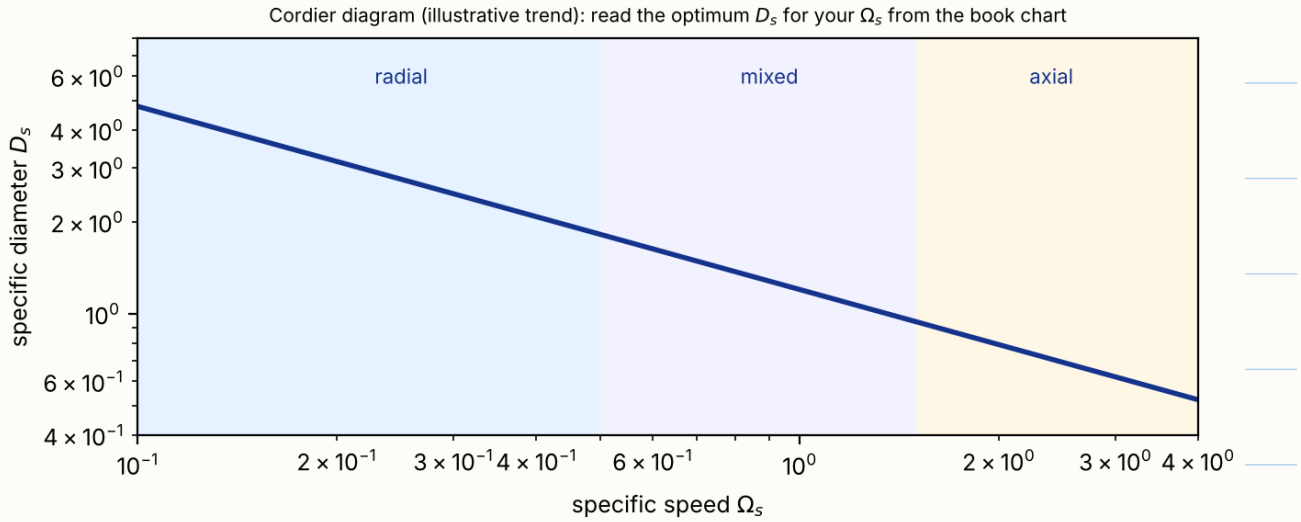
Example: *A pump runs at 1450 rpm and gives a head of 20 m.*

Estimate the head at 2900 rpm with the same diameter.

Solution: $H_2 = 20 \times (2900/1450)^2 = 20 \times 4 = 80$ m.

4. Specific speed

Cordier diagram with radial, mixed and axial zones



Specific speed is a dimensionless parameter that describes the shape of a machine. It depends only on operating point (flow rate and head) and not on size. It is used to choose the machine type.

$$\Omega_s = \omega \sqrt{Q} / (gH)^{3/4} \quad (\omega \text{ in rad/s, } Q \text{ in m}^3/\text{s, } H \text{ in m})$$

Specific diameter is the companion parameter:

$$D_s = D\,(gH)^{1/4} / \sqrt{Q}$$

- Low Ω_s (about 0.1 to 0.5): radial (centrifugal) machine, high head, low flow.
- Medium Ω_s (about 0.5 to 1.5): mixed flow machine.
- High Ω_s (above about 1.5): axial flow machine, high flow, low head.

Example: *A pump has Q = 0.1 m³/s, H = 30 m and N = 1450 rpm. Find the specific speed and the machine type.*

Solution: $\omega = 2\pi \times 1450/60 = 151.9$ rad/s. $\sqrt{Q} = 0.316$. $(gH)^{3/4} = (9.81 \times 30)^{0.75} = 71.0$. So $\Omega_s = 151.9 \times 0.316 / 71.0 = 0.68$. This is a mixed flow pump.

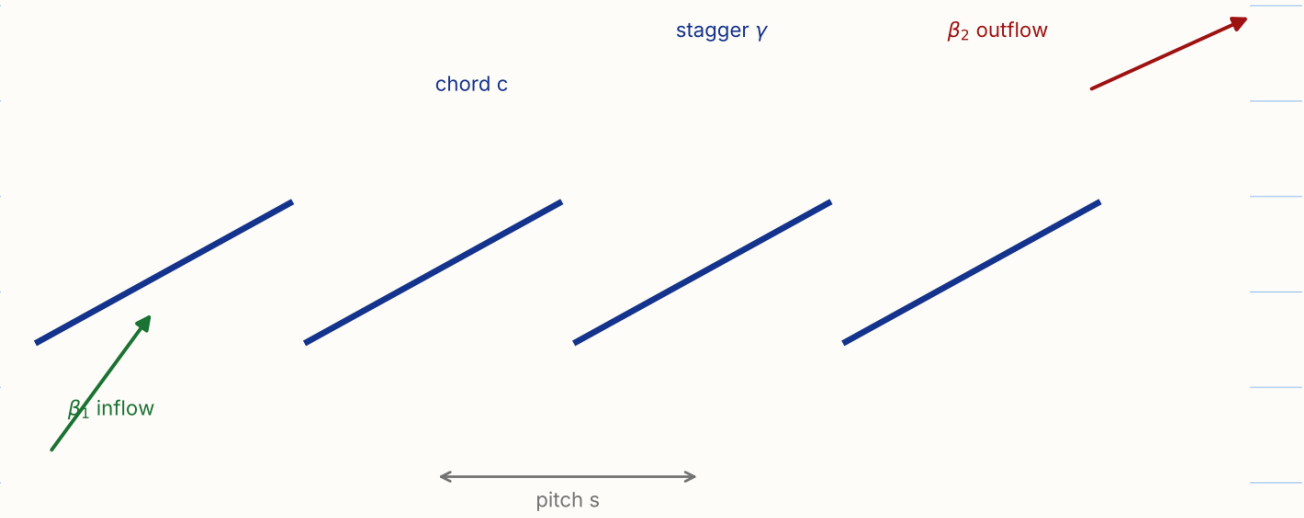
5. Cordier diagram

Cordier plotted the optimum (maximum efficiency) specific diameter D_s against specific speed Ω_s for a large set of real machines. The data collapse onto a single curve. The main message is:

- Low Ω_s needs a large D_s, which is why radial machines have large diameter and small flow.
- High Ω_s needs a small D_s, which is why axial machines are slim and fast.
- The Cordier diagram is used at the preliminary design stage to select the type and approximate size of a machine.

6. Two-dimensional cascade: definitions

Cascade: pitch, chord, stagger and flow angles



A cascade is a row of blades laid out on a plane, as if the blades were unrolled from a stage at a given radius. The main geometric quantities are:

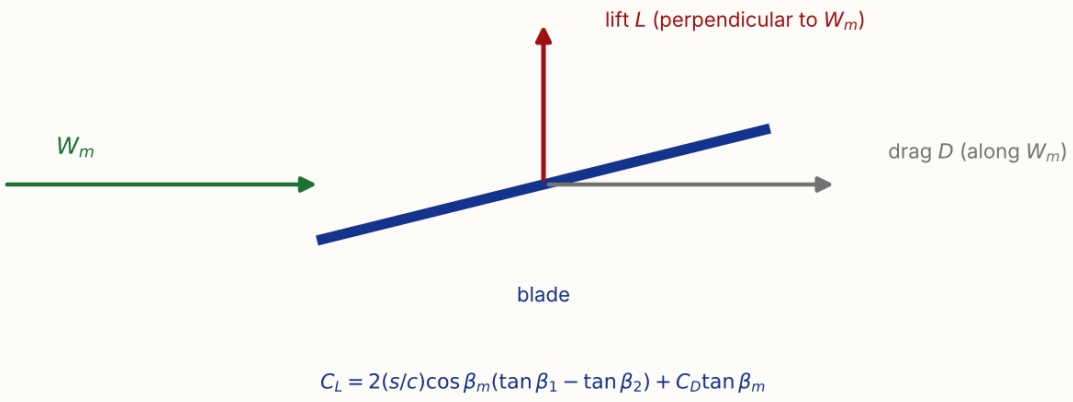
- **Pitch s:** spacing between blades. **Chord c:** length of blade from leading edge to trailing edge.

- **Solidity σ = c / s.** Stagger angle γ: angle of the chord to the axial direction.
- **Inlet flow angle β_1** and **outlet flow angle β_2**: angles of the relative flow measured from axial.
- **Incidence i** = $\beta_1 - \beta_1'$ (difference between actual and blade inlet angle). **Deviation δ** = $\beta_2 - \beta_2'$.

Camber θ = $\beta_1 - \beta_2$ **Mean flow angle tan β_m = ½ (tan β_1 + tan β_2)**

7. Lift and drag in a cascade

Lift and drag on a blade in a cascade



The cascade turns the flow by θ. The change in tangential velocity across the cascade produces a force on the blade. This force can be

split into lift (perpendicular to the mean velocity) and drag (parallel to it).

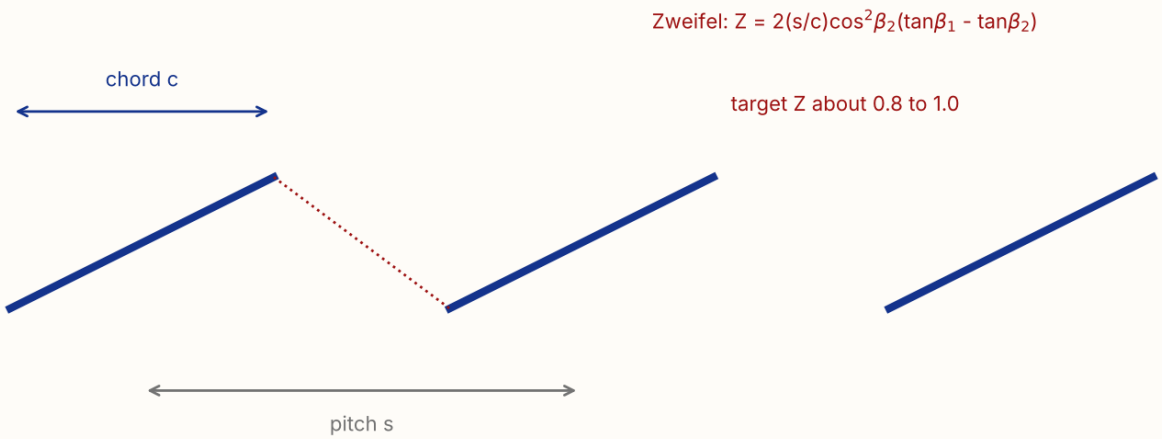
$$C_L = 2\,(s/c)\cos\beta_m\,(\tan\beta_1 - \tan\beta_2) + C_D \tan\beta_m$$

Here C_D is the profile drag coefficient. The first term is the lift that does the turning. The second term is the drag contribution to the force along the axial direction. For good cascades C_D is small compared with C_L, so the lift term dominates.

Lift-to-drag ratio is a measure of cascade quality. Good designs have a high C_L/C_D, which gives high blade efficiency.

8. Zweifel criterion (blade loading)

Zweifel loading: pitch against chord



Zweifel proposed a rule for the best pitch. The loading coefficient should be about 0.8 to 1.0 for minimum loss:

$$Z = 2\,(s/c)\cos^2\beta_2\,(\tan\beta_1 - \tan\beta_2) \quad \text{target } Z \approx 0.8 \text{ to } 1.0$$

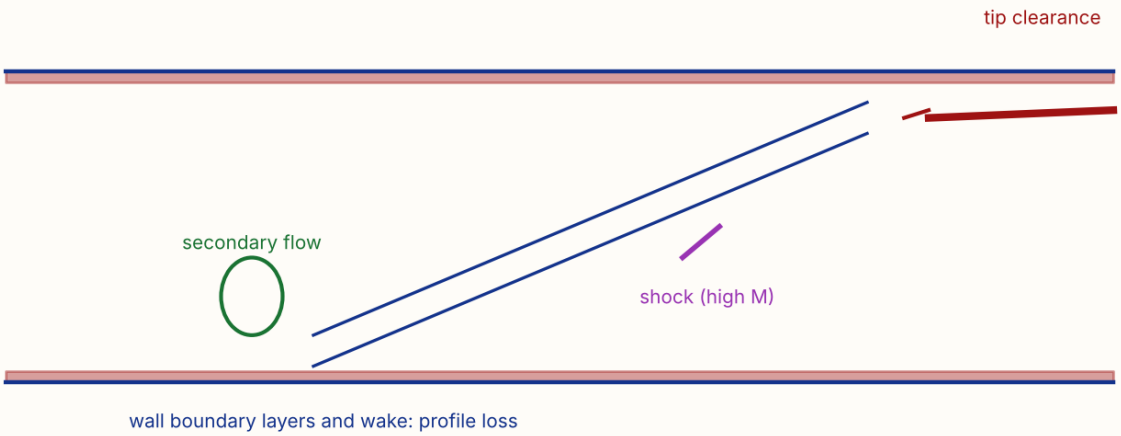
Example: *For a compressor cascade with $\beta_1 = 60°$, $\beta_2 = 30°$ and s/c = 0.6, check the Zweifel number.*

Solution: $\cos^2 30° = 0.75$, $\tan 60° - \tan 30° = 1.732 - 0.577 = 1.155$.

$Z = 2 \times 0.6 \times 0.75 \times 1.155 = 1.04$. This is slightly high, so a slightly larger s/c would reduce loading, at the cost of more blades.

9. Blade efficiency and loss estimation

Cascade passage and loss sources



Losses in a cascade are measured by the pressure loss coefficient:

$$\zeta = (p_{01} - p_{02}) / (\tfrac{1}{2} \rho W_1^2)$$

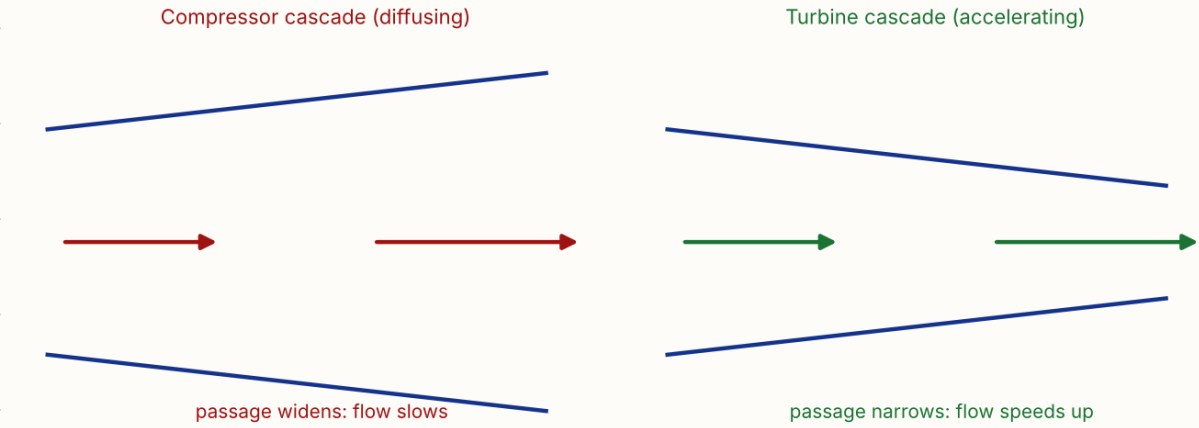
Losses have several sources, and they are added to give the total:

- **Profile loss:** boundary layer growth and wake on the blade surface.
- **Secondary flow loss:** passage vortices and endwall boundary layers.
- **Tip clearance loss:** leakage over the blade tip.
- **Shock loss:** at high Mach number on the blade surface (compressor blades at high relative Mach).

Blade efficiency for a compressor cascade is roughly $1 - (C_D/C_L) \times$ (something of order 1), so high lift and low drag mean high efficiency. Empirical correlations (for example, Ainley and Mathieson) estimate ζ from geometry and flow angle.

10. Compressor and turbine cascades

Diffusing (compressor) vs accelerating (turbine) passage



Item	Compressor cascade	Turbine cascade
Flow	Decelerating (diffusing)	Accelerating (expanding)
Main loss risk	Separation, stall	Mostly profile and trailing-edge loss
Typical loading	Moderate (Zweifel near 0.8–1.0)	Higher turning possible
Key limit	Diffusion factor	Exit Mach and trailing edge thickness

Compressor cascades are limited by diffusion, because the flow must slow down and the boundary layer can separate. Turbine cascades accelerate the flow, so they can be loaded more heavily without separation.

11. Worked example

Example: *Find the mean flow angle for $\beta_1 = 60°$ and $\beta_2 = 30°$.*

Solution: $\tan \beta_m = ½ (1.732 + 0.577) = 1.1547$, so $\beta_m = 49.1°$.

12. Exam quick points

- Similarity: keep φ, ψ, Re, M the same. Affinity: $Q \propto ND^3$, $H \propto N^2D^2$, $P \propto N^3D^5$.
- Ω_s tells us the machine type. Cordier diagram gives D_s for best efficiency.
- Cascade: s/c, solidity, camber, deviation, incidence. Zweifel target about 0.8 to 1.0.
- Losses: profile, secondary, tip clearance, shock.

13. Practice questions

Q1. Define specific speed and explain how it is used to select a pump type.

Q2. State the affinity laws and use them to find the flow rate when speed is doubled at constant diameter.

Q3. Explain what the Zweifel coefficient represents and what happens if the pitch is too large or too small.

UNIT 4: Axial Compressor; Centrifugal Pump and Compressor

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Syllabus covered: Two-dimensional pitch-line design and analysis; h-s diagram; degree of reaction; Mach number effect; performance and efficiency; three-dimensional flow; tip clearance; losses; theoretical analysis of centrifugal pump and compressor; circulation and Coriolis effects; reversed eddies; slip factor; head and efficiency; diffuser; combustion system introduction.

A. Axial compressor

1. Two-dimensional pitch-line design

Pitch-line design works at the mean radius of the blade. Velocity triangles are drawn at inlet and outlet of the rotor and stator. The usual assumptions are: axial velocity C_x is constant through the stage, and the radius at the blade root and tip is the same as at the pitch line.

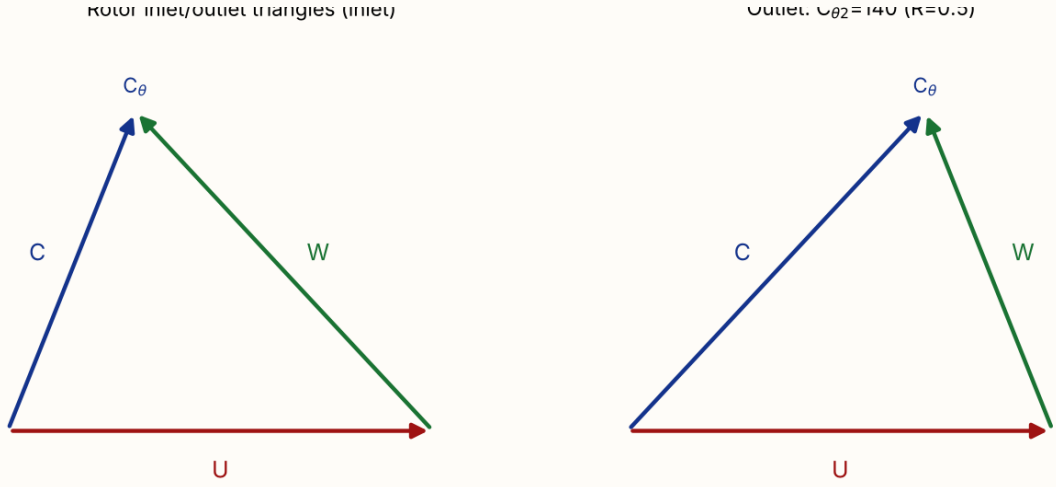
Stage work $\Delta h_0 = U\,(C_{\theta 2} - C_{\theta 1})$ **Stage loading $\psi = \Delta h_0 / U^2$**

Flow coefficient $\varphi = C_x / U$

The design process is: choose U and C_x from the required flow and pressure rise, choose the stage loading, then find the blade angles from the velocity triangles.

2. Degree of reaction

Velocity triangles for 50% reaction



The degree of reaction R is the fraction of the stage enthalpy rise that happens in the rotor:

$$\mathbf{R = \Delta h_{rotor} / \Delta h_{stage} \quad \Rightarrow \quad R = 1 - (C_{\theta 1} + C_{\theta 2}) / (2U)}$$

(constant C_x)

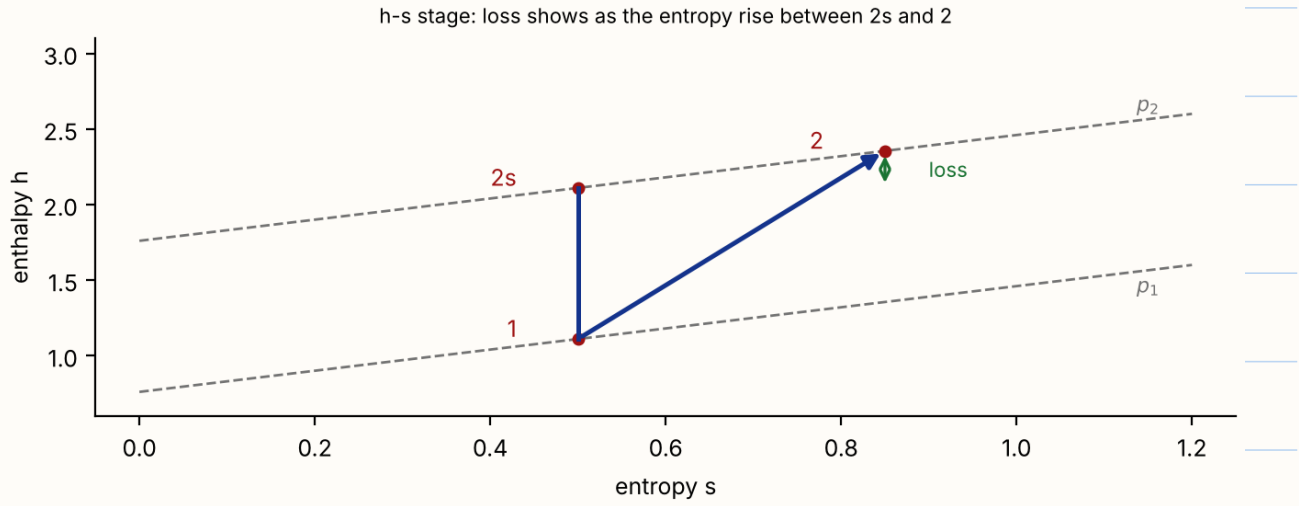
- R = 0: all the rise is in the stator (impulse-like design).
- R = 0.5: half the rise in rotor and half in stator. Symmetric velocity triangles, widely used.
- R = 1: all the rise in the rotor.

Example: *U = 200 m/s, C_x = 150 m/s, $C_{\theta 1}$ = 60 m/s. Find $C_{\theta 2}$ for 50% reaction, and the stage loading.*

Solution: R = 0.5 gives $C_{\theta 1} + C_{\theta 2} = U = 200$, so $C_{\theta 2} = 140$ m/s. $\Delta h_0 = 200 \times (140 - 60) = 16$ kJ/kg. $\psi = 16000 / 200^2 = 0.4$.

3. The h-s diagram

h-s stage with the loss gap marked



On the h-s (enthalpy-entropy) diagram, the stage is shown as a path from state 1 to state 2. The isentropic path gives Δh_{0s}, while the actual path gives Δh_0. The vertical gap between the two paths at the same pressure shows the loss. Entropy rise is caused by losses.

To sketch: draw two isobars p_{01} and p_{02}, draw the isentropic line (vertical on h-s), and then draw the actual path to the right of it. Mark the stagnation enthalpy rise Δh_0. Stage efficiency is $\eta = \Delta h_{0s} / \Delta h_0$.

4. Effect of Mach number

At the blade tip the relative velocity W is highest, because U is largest. If W reaches supersonic speed, a shock forms on the blade surface. Shocks increase losses, reduce pressure rise and can cause stall. Typical design limits are a relative tip Mach number of about 1.1 to 1.3.

Relative tip Mach number $M_{w,tip} = W_{1,tip} / a_1$

5. Diffusion factor and stall

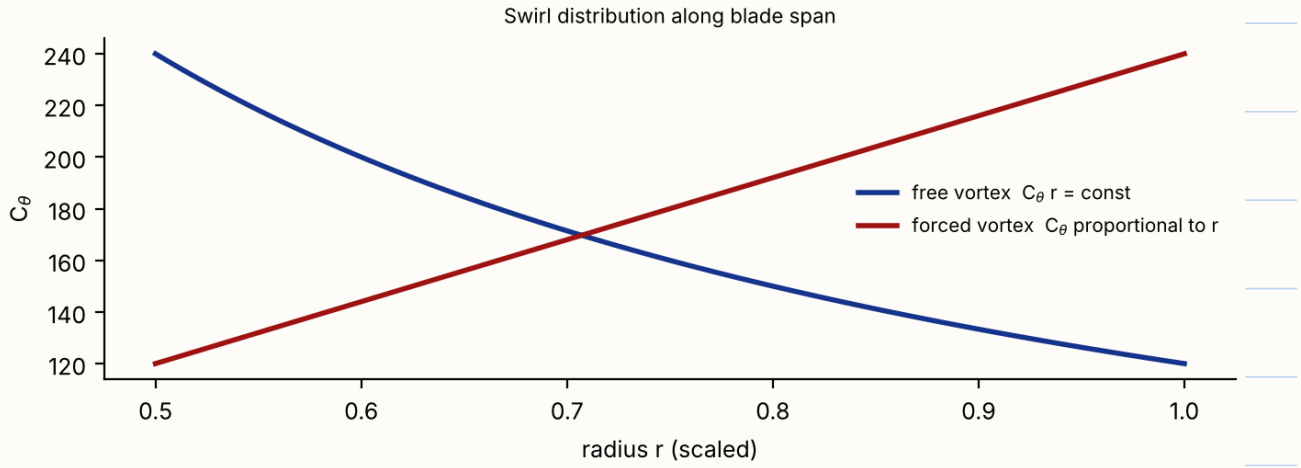
The Lieblein diffusion factor measures how much the blade row slows the relative flow. It is used to predict separation:

$DF = 1 - W_2/W_1 + \Delta C_\theta / (2\sigma W_1)$ σ = solidity = c/s

A diffusion factor below about 0.6 is generally considered safe. Higher values indicate a risk of separation and stall.

6. Three-dimensional flow and radial equilibrium

Free vortex vs forced vortex



The flow is not purely two-dimensional. Along the span there is a radial pressure gradient that balances the centrifugal force on the fluid. This is radial equilibrium:

$dp/dr = \rho C_\theta^2 / r$ (simplified radial equilibrium)

- **Free vortex design:** C_θ r = constant. Stage work is then the same at all radii, but the blade loading changes along the span.
- **Forced vortex:** C_θ proportional to r. Used in some rotors at low work.
- Designers often choose free vortex for the rotor to keep work uniform along span.

7. Tip clearance

A gap exists between the rotor tip and casing. Pressure difference across the blade drives fluid through the gap (tip leakage). The leakage flow

reduces work done on the fluid, produces a leakage vortex and increases loss. Efficiency falls roughly linearly as clearance increases. Keeping clearance small is therefore important.

8. Losses and performance

- Profile loss, secondary (endwall) loss, tip clearance loss, shock loss, and incidence loss.
- Performance is shown on a compressor map: pressure ratio against corrected mass flow, with lines of constant speed and efficiency islands.
- **Surge:** flow reverses in the compressor at low mass flow; the surge line bounds the stable range.
- **Stall:** local separation on blades, can grow into rotating stall. **Choke:** maximum flow when a passage reaches sonic speed.

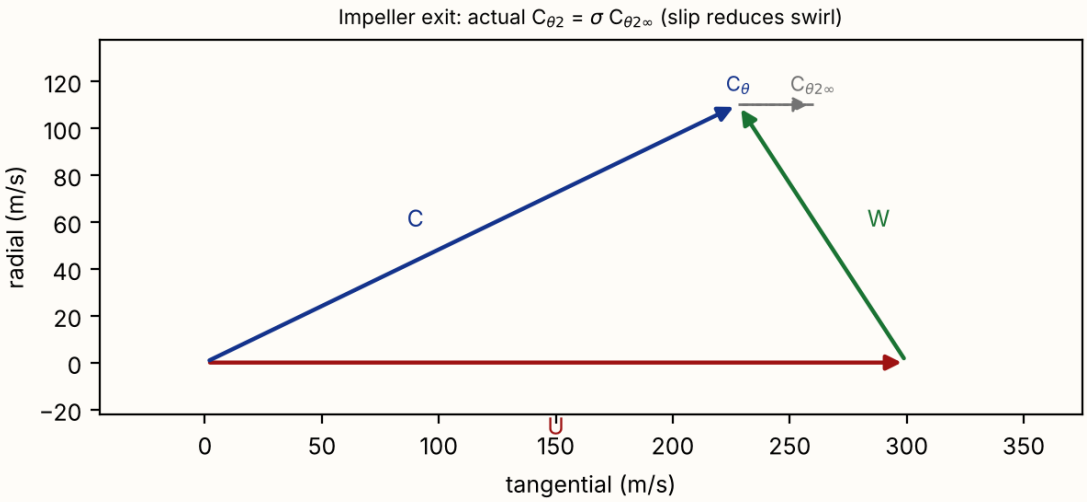
Quantity	Symbol	Meaning
Stage loading	ψ	$\Delta h_0 / U^2$, how hard the stage works
Flow coefficient	φ	C_x / U, how much flow passes
Reaction	R	Fraction of rise in rotor

Diffusion factor	DF	Predicts separation (safe below 0.6)

B. Centrifugal pump and compressor

9. Theoretical head and Euler equation

Impeller exit triangle with slip



In a centrifugal impeller, the fluid enters axially and leaves radially. The Euler head is:

$H_E = (U_2\, C_{\theta 2} - U_1\, C_{\theta 1}) / g$ with no pre-whirl ($C_{\theta 1} = 0$): $H_\infty = U_2 C_{\theta 2\infty} / g$

For an impeller with infinite blades (ideal), the exit relative flow follows the blade angle exactly. Blade angle relates the exit tangential velocity to the radial velocity:

$$C_{\theta 2\infty} = U_2 - C_{r2} \cot \beta_2 \quad \Rightarrow \quad H_\infty = (U_2^2 - U_2 C_{r2} \cot \beta_2) / g$$

10. Slip factor

With a finite number of blades, the actual exit flow turns less than the blade angle. The fluid slips. The slip factor σ is the ratio of actual to ideal tangential velocity:

$$\sigma = C_{\theta 2} / C_{\theta 2\infty} \quad \text{Stanitz: } \sigma = 1 - 0.63\pi / Z \quad \text{Wiesner: } \sigma = 1 - \sqrt{(\sin \beta_2)} / Z^{0.7}$$

Z is the number of blades. More blades give less slip. Real head is H = σ U_2 $C_{\theta 2\infty}$ / g minus losses.

Example: *A centrifugal impeller has Z = 16 and $\beta_2 = 40°$. Find the slip factor using Stanitz's formula, with $U_2 = 300$ m/s and $C_{\theta 2\infty} = 260$ m/s.*

Solution: $\sigma = 1 - 0.63\pi/16 = 1 - 0.124 = 0.876$. Actual $C_{\theta 2} = \sigma \times C_{\theta 2\infty} = 0.876 \times 260 = 228$ m/s. Ideal Euler head $H_\infty = U_2 C_{\theta 2\infty}/g = 300 \times 260 / 9.81 = 7950$ m of fluid. Slip reduces the head to about 0.876 of this, so H ≈ 6960 m of fluid.

11. Circulation, Coriolis and reversed eddies

The Coriolis effect and blade circulation produce a non-uniform relative velocity inside the impeller passage. Low-velocity, high-loss fluid collects in one part of the passage. Near the exit this distorts the flow and increases slip.

Reversed eddies: at low flow rates (below the design flow), the incidence on the inlet becomes large. Flow separates and recirculates near the inlet or near the exit. This reduces efficiency and causes unstable operation. Designers avoid operation in this region.

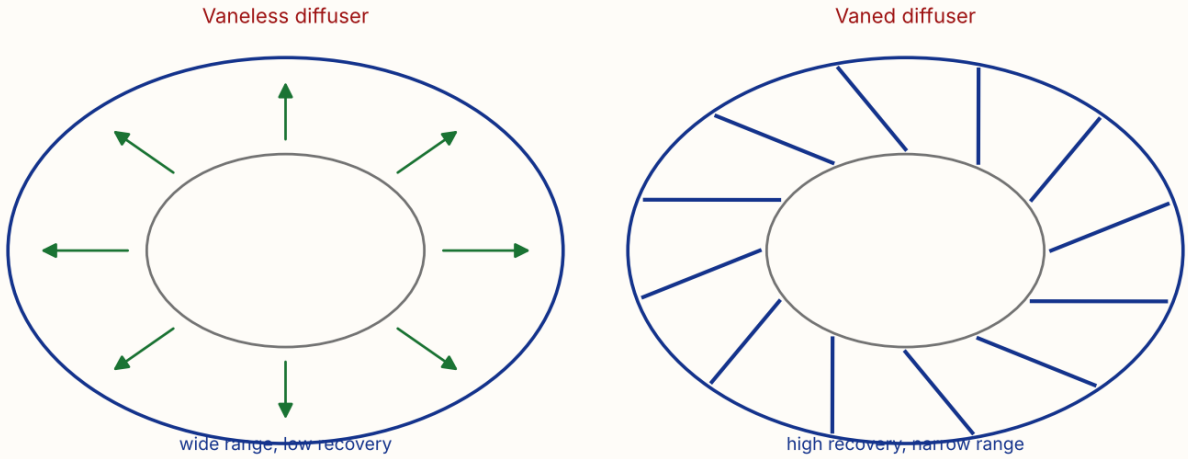
12. Head and efficiency

Hydraulic efficiency η_h = actual head / Euler head Overall η = (ρ g Q H) / shaft power

Losses in a centrifugal machine include disc friction, leakage, incidence, skin friction in the passage, and diffuser losses.

13. Diffuser

Vaneless and vaned diffusers



The diffuser recovers pressure from the high kinetic energy at the impeller exit. Two types are common:

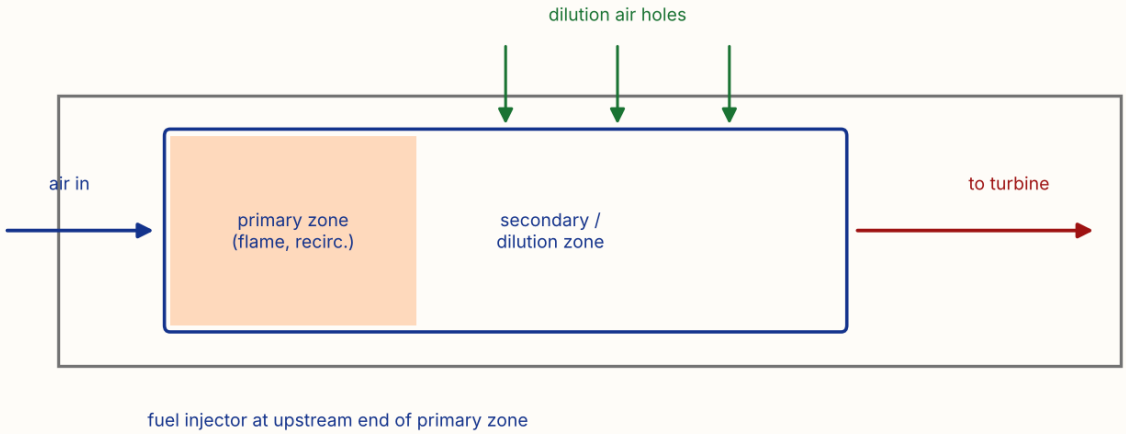
- **Vaneless diffuser:** simple, wide operating range, low pressure recovery.
- **Vaned diffuser:** higher pressure recovery, narrower operating range, more loss at off-design.

Pressure recovery coefficient $C_p = (p_2 - p_1) / (\frac{1}{2} \rho C_1^2)$

ideal: $C_{p,ideal} = 1 - (A_1/A_2)^2$

14. Introduction to the combustion system

Combustor cross-section



The combustor adds heat to compressed air by burning fuel. It must burn fuel completely, keep the pressure loss low (typically 3–6%), and produce a uniform exit temperature for the turbine. Common types are can, annular and can-annular combustors. Inside the combustor there is a primary zone (rich burning and recirculation for flame stability), a secondary zone (more air) and a dilution zone (cooling air to the turbine temperature).

Equivalence ratio $\varphi = (F/A)_{actual} / (F/A)_{stoich}$ (kerosene stoichiometric F/A ≈ 0.068, air-fuel ≈ 15)

15. Exam quick points

- Axial: $\Delta h_0 = U\Delta C_\theta$, $R = 1 - (C_{\theta 1}+C_{\theta 2})/(2U)$. Keep DF below 0.6.
- Shock at tip if $M_{w,tip}$ is too high. Tip clearance reduces efficiency.

- Centrifugal: $H = U_2 C_{\theta 2}/g$. Slip factor $\sigma < 1$, Stanitz and Wiesner.
- Reversed eddies at low flow. Vaneless diffuser for range, vaned for recovery.

16. Practice questions

Q1. Derive the expression for degree of reaction for constant axial velocity.

Q2. Explain the difference between free vortex and forced vortex designs and why the free vortex is preferred for rotors.

Q3. Explain slip factor. A centrifugal impeller has 12 blades, $\beta_2 = 35°$. Find σ using Wiesner's relation ($\sin 35° = 0.574$, $Z^{0.7} = 12^{0.7} = 5.28$).

Q4. Describe the function of the combustor primary zone and dilution zone.

UNIT 5: Axial Flow Turbine and CFD for Turbomachinery

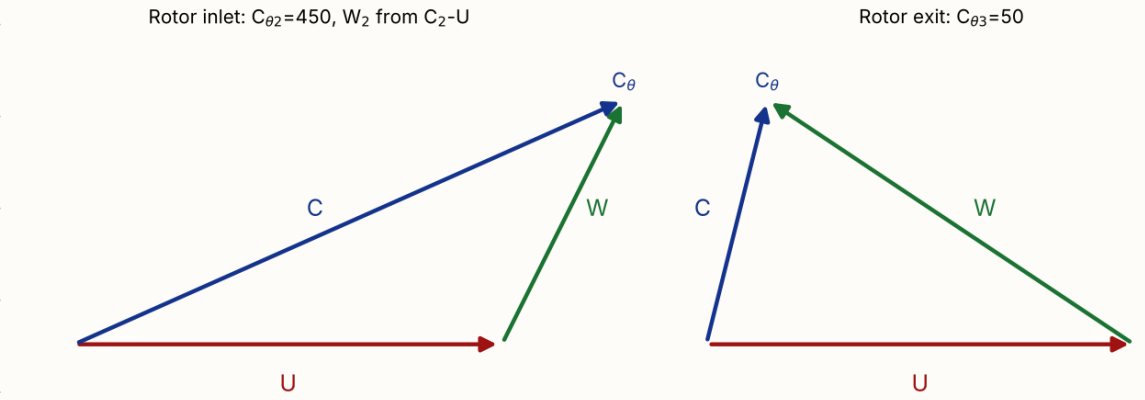
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Syllabus covered: Two-dimensional pitch-line design; stage loading capacity; degree of reaction; stage efficiency; turbine performance; blade cooling; examples. CFD: governing equations, numerical methods, test cases for flow and heat transfer.

A. Axial flow turbine

1. Two-dimensional pitch-line design

Turbine rotor inlet and exit triangles



The axial turbine has a stator (nozzle) row that accelerates the flow and turns it, followed by a rotor row that extracts work. The pitch-line design

uses velocity triangles at the mean radius, with C_x constant through the stage.

Stage work $\Delta h_0 = U\,(C_{\theta 2} - C_{\theta 3})$ **Stage loading $\psi = \Delta h_0 / U^2$**

Flow coefficient $\varphi = C_x / U$

Here station 2 is the nozzle exit (rotor inlet) and station 3 is the rotor exit. C_θ is measured in the direction of rotation. The sign convention must be stated clearly in the exam, because the turbine has opposite sign to the compressor.

2. Stage loading capacity

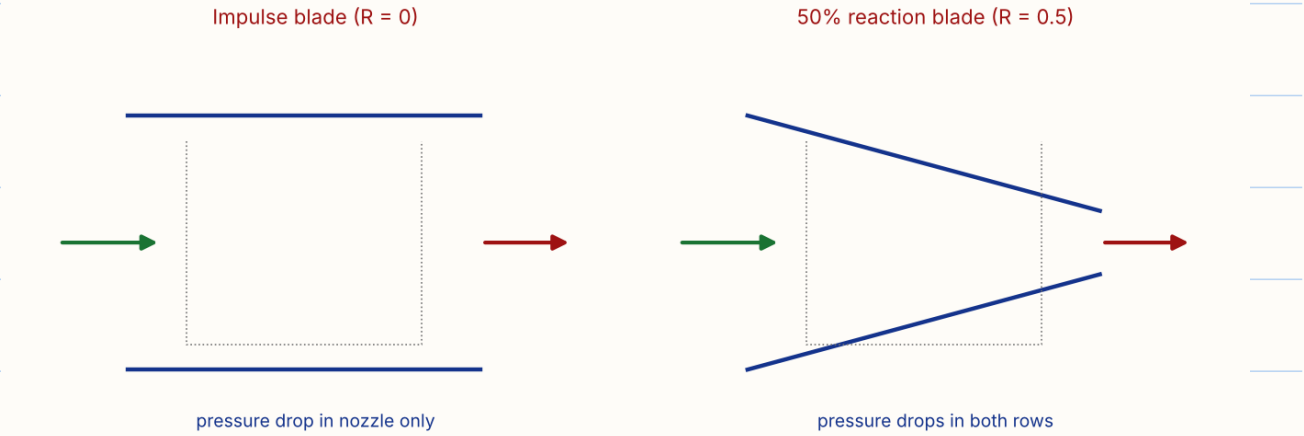
Stage loading capacity is how much work one stage can extract for a given blade speed. For high loading, the flow must turn a lot. Turbine stages can be loaded more heavily than compressor stages because the flow accelerates and separation is less of a risk. Typical ψ for a turbine stage is about 1.0 to 2.5 (higher than for compressors).

Example: *A turbine stage has U = 350 m/s, $C_{\theta 2}$ = 450 m/s and $C_{\theta 3}$ = 50 m/s. Find Δh_0 and the stage loading.*

Solution: $\Delta h_0 = 350 \times (450 - 50) = 140$ kJ/kg. $\psi = 140\,000 / 350^2 = 140\,000 / 122\,500 = 1.14$.

3. Degree of reaction and stage efficiency

Impulse and 50% reaction blades



Degree of reaction for a turbine is the fraction of the stage enthalpy drop that happens in the rotor:

$$R = \Delta h_{rotor} / \Delta h_{stage} = (W_2^2 - W_3^2) / (2\,U\,(C_{\theta 2} - C_{\theta 3}))$$

- **Impulse stage** (R ≈ 0): the expansion happens in the nozzle only. The rotor just turns the flow.
- **50% reaction stage** (R = 0.5): equal expansion in nozzle and rotor, symmetric blades, widely used.
- Higher reaction gives more expansion in rotor, which needs more blade tip clearance and gives higher stage loading in the rotor.

Stage efficiency compares the actual stage work with the isentropic work:

$\eta_{stage} = \Delta h_0 / \Delta h_{0s}$ (total-to-total or total-to-static, state which one)

4. Turbine performance

Turbine performance is shown on a characteristic map. The key variables are corrected mass flow, pressure ratio, speed and efficiency. Important features:

- The nozzle chokes when the pressure ratio is high enough for sonic flow at the throat. Beyond this point mass flow is fixed by the nozzle throat area.
- Turbine mass flow depends on inlet pressure and temperature. Corrected mass flow is used to compare operation at different inlet conditions.
- Off-design: at lower speed or pressure ratio, incidence changes and the efficiency falls.
- Losses: profile, secondary flow, tip leakage (in the rotor), trailing edge loss and cooling-air mixing losses.

Parameter	Effect on turbine	Remark

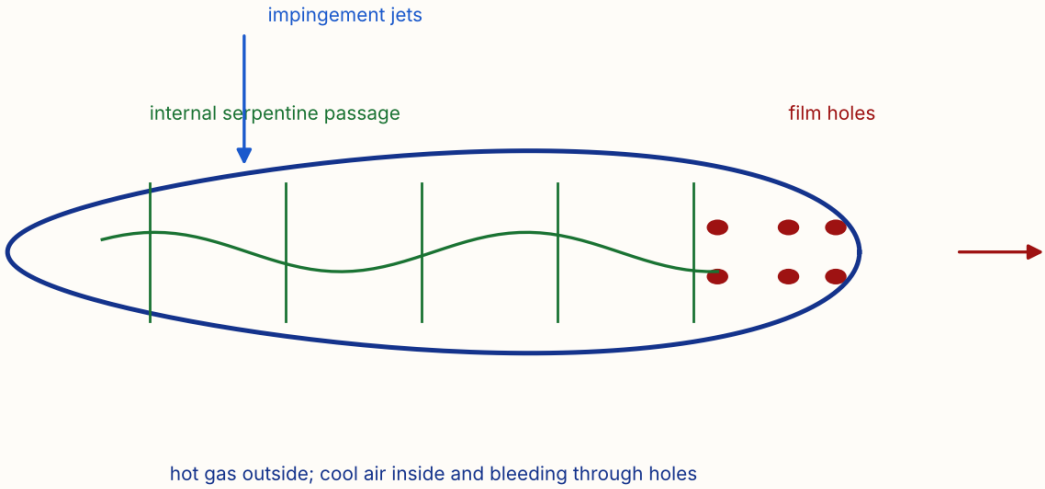
Higher pressure ratio	More work, nozzle may choke	Limit set by choking
Higher speed	Higher U, changes incidence	Efficiency peaks near design speed
Cooling air bleed	Less mass flow through rotor	Reduces cycle efficiency

5. Why blade cooling is needed

Turbine inlet temperature in modern engines is 1600 to 1800 K. Blade metal can handle only about 1100 to 1200 K. Cooling is essential to keep the blades alive. Higher inlet temperature increases specific work and cycle efficiency, so cooling technology allows a higher temperature to be used.

6. Blade cooling methods

Blade cooling methods



- **Internal convection cooling:** air flows through serpentine passages inside the blade and removes heat from the inner surface.
- **Impingement cooling:** jets of cool air strike the inner wall of the blade leading edge at high heat transfer rates.
- **Film cooling:** cool air exits through holes on the blade surface and forms a protective layer over the outer surface.
- **Transpiration cooling:** air passes through porous walls. Gives uniform film cover but is hard to manufacture.
- **Thermal barrier coating (TBC):** ceramic layer on the blade surface that insulates the metal.

Film cooling effectiveness $\eta_{fc} = (T_g - T_{aw}) / (T_g - T_c)$

Example: *Hot gas is at T_g = 1700 K, coolant at T_c = 800 K and effectiveness η_{fc} = 0.5. Find the adiabatic wall temperature T_{aw}.*

Solution: $T_{aw} = T_g - \eta_{fc}(T_g - T_c) = 1700 - 0.5 \times 900 = 1250$ K. This is above the metal limit, so more cooling or a TBC is needed.

Cost of cooling: cooling air is taken from the compressor and does not produce work in the turbine, so it reduces cycle efficiency. Designers balance higher inlet temperature against cooling air requirement.

B. CFD applied to turbomachinery

7. Governing equations

CFD solves the conservation equations for mass, momentum and energy. For a compressible viscous flow in an inertial (stationary) frame, in conservative form:

Continuity: $\partial\rho/\partial t + \nabla\cdot(\rho \mathbf{V}) = 0$ **Momentum:** $\partial(\rho\mathbf{V})/\partial t + \nabla\cdot(\rho\mathbf{V}\mathbf{V}) = -\nabla p + \nabla\cdot\tau + \mathbf{S_M}$ **Energy:** $\partial(\rho E)/\partial t + \nabla\cdot(\mathbf{V}(\rho E + p)) = \nabla\cdot(k\nabla T) + \nabla\cdot(\tau\cdot\mathbf{V})$

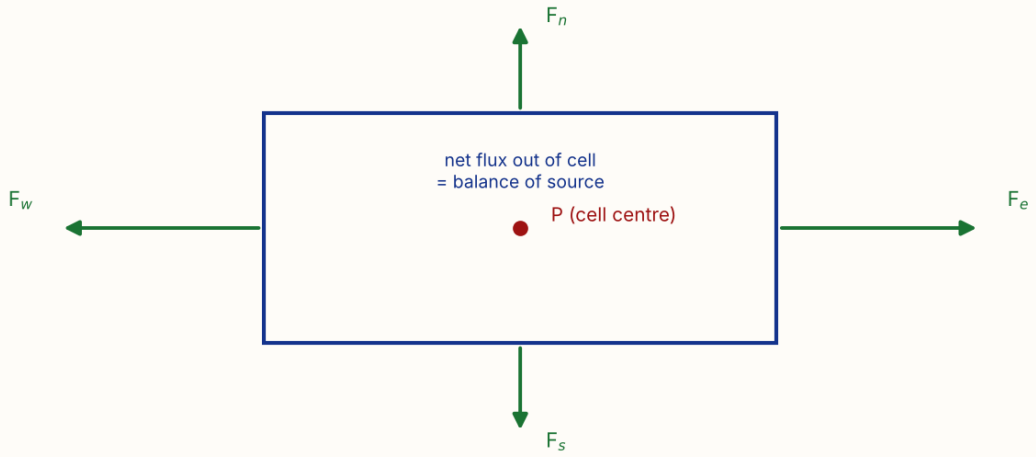
In a rotating blade row it is common to write the equations in the rotating frame. Extra source terms appear: the Coriolis term $-2\rho(\Omega \times W)$ and the

centrifugal term $-\rho\Omega \times (\Omega \times r)$. These are the same terms derived in Unit 1.

Turbulence: real turbomachinery flows are turbulent. Reynolds-averaged Navier-Stokes (RANS) equations are used, with a turbulence model. Common models are the k-ε model (good in free-stream) and the k-ω SST model (better near walls and for separation). LES and DNS are used for research because they are expensive.

8. Numerical methods

Finite volume cell



- **Finite volume method:** most common in turbomachinery CFD. The domain is divided into control volumes, and conservation is enforced on each one.
- **Finite difference method:** used for structured grids, simple geometry and research codes.

- **Finite element method:** more common in structural analysis, sometimes used for heat transfer.
- **Discretisation schemes:** upwind (stable, diffusive) and central differences (accurate, can oscillate). Second-order schemes are standard for final results.

Time marching: steady-state problems are solved by iterating in pseudo-time until residuals drop. Unsteady problems are solved with time-accurate methods. Explicit schemes need a small time step, controlled by the CFL number:

CFL = u Δt / Δx ≤ 1 (for explicit schemes, stability limit)

Example: *Grid spacing Δx = 0.1 mm and velocity u = 200 m/s. Find the maximum time step for CFL = 1.*

Solution: $\Delta t \leq \Delta x / u = 0.1 \times 10^{-3} / 200 = 0.5$ μs.

9. Grids and modelling of rotating blades

Structured grids (H-grid, O-grid, C-grid) are popular for blades because they give good control of wall resolution. **Unstructured grids** are flexible for complex geometry. Near-wall resolution is measured by y+:

$y+ = u_\tau\, y / \nu$ ($y+ \approx 1$ for wall-resolved; $y+ \approx 30$ to 100 for wall-function models)

Stage interaction between rotor and stator is handled by a **mixing plane** (circumferential average at the interface) or a **sliding mesh** (rotor moves relative to stator, unsteady). Frozen rotor is a steady approximation that fixes the relative position.

10. Test cases

Test cases are used to check the solver and the model before using it for design. Good test cases for turbomachinery flow and heat transfer are:

- **Flat plate boundary layer:** checks skin friction and heat transfer (Stanton number) against correlations.
- **Turbine or compressor cascade:** checks pressure distribution, loss coefficient and flow angles against measured data.
- **Single stage or multi-stage test rig:** checks total pressure ratio, efficiency and mass flow over a range of speeds.
- **Blade heat transfer:** checks heat transfer coefficient on the blade surface, including film-cooled cases.

Results are validated by comparing with experimental data and by grid-independence studies (refining the grid until the solution no longer changes).

11. Exam quick points

- Turbine: $\Delta h_0 = U(C_{\theta 2} - C_{\theta 3})$. Stage loading ψ can be 1 to 2.5.
- Reaction $R = \Delta h_{rotor} / \Delta h_{stage}$. Impulse $R \approx 0$, 50% reaction $R = 0.5$.
- Blade cooling: internal convection, impingement, film cooling, TBC. $\eta_{fc} = (T_g - T_{aw}) / (T_g - T_c)$.
- CFD: continuity, momentum, energy. RANS with k-ω SST. Finite volume most common. CFL ≤ 1 for explicit. y+ controls near-wall grid.

12. Practice questions

Q1. Draw a velocity triangle for a 50% reaction turbine stage and explain why the stator and rotor have symmetric blades.

Q2. Explain the difference between film cooling and impingement cooling, and when each is used.

Q3. A turbine stage has $\Delta h_0 = 120$ kJ/kg and U = 300 m/s. Find the stage loading.

Q4. State the three conservation equations solved in CFD and explain why RANS is used for turbomachinery design.